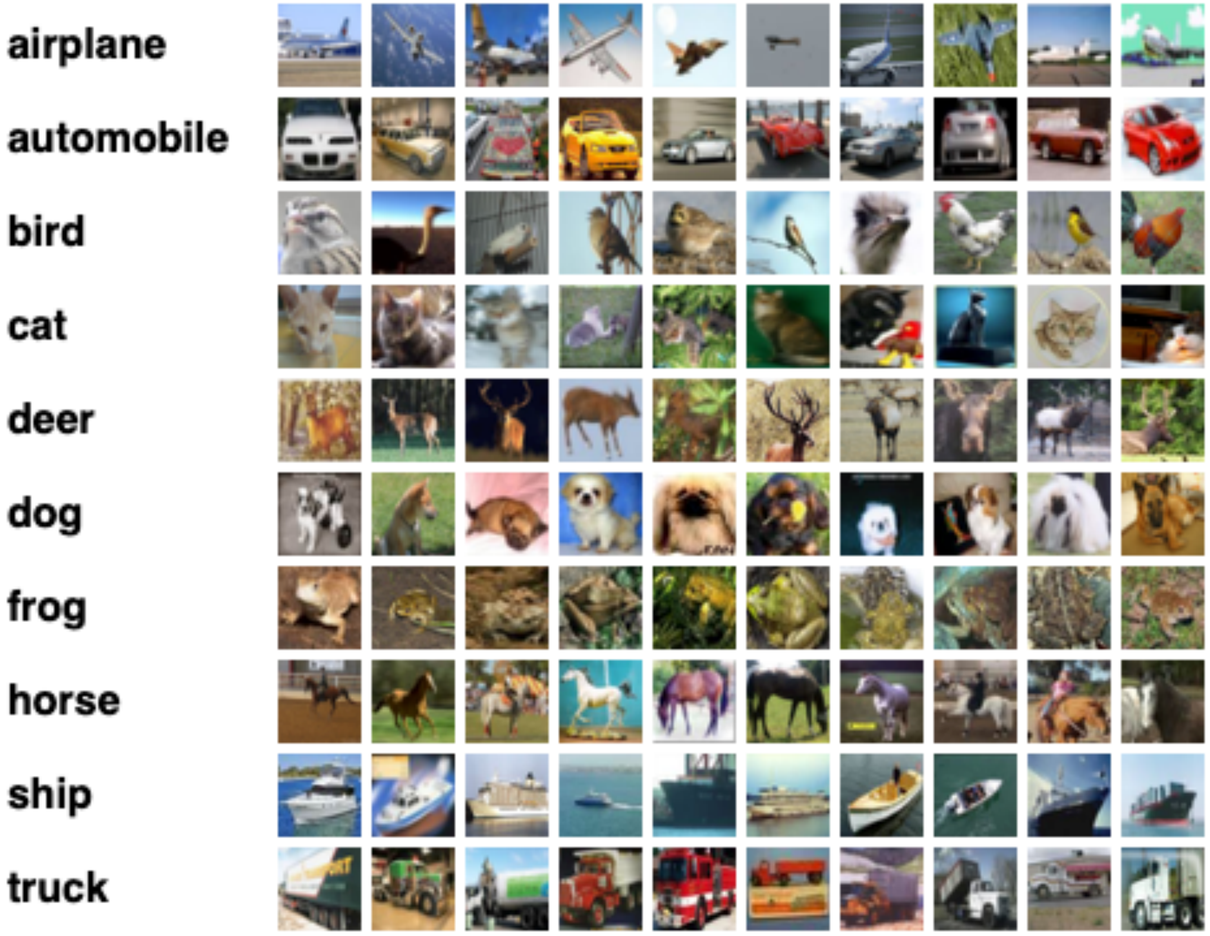
Capstone 9.5 - Deep Learning

Image Classification with CNN using CIFAR-10 Dataset

Data - CIFAR-10

The CIFAR-10 dataset consists of 60000 32x32 color images in 10 classes, with 6000 images per class. There are 50000 training images and 10000 test images.

The classes are completely mutually exclusive. There is no overlap between automobiles and trucks. "Automobile" includes sedans, SUVs, things of that sort. "Truck" includes only big trucks. Neither includes pickup trucks.



Task A
Loading and Preparing the Data

- Import necessary libraries and load the CIFAR-10 dataset
- Import Keras and its layers
- Load the CIFAR-10 dataset using `keras.datasets.cifar10.load_data()`
  - Hint: similar to MNIST data load
  - Make sure you split the data into training and testing sets
- Inspect the data shape for all subsets
- Plot a bar chart to visualize the label distribution
- Normalize the data using the division method

  Hint: `classes = ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']`

Task B
Building, Training, and Evaluating the CNN

- Create a CNN sequential model using the `add()` method
  - Build at least 2 convolutional layers
  - Add Dropout to one of the layers
- Compile the model with appropriate components
- Train the model and save history
- Evaluation
  - Testing data using the evaluate function
  - Build accuracy and loss graphs
  - Visual inspection for a sample of images
    - Hint: `classes = ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']`